

Hardware-Aware Simulation of Organic Optoelectronic Compute-in-Memory Inference for Edge Vision

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Abstract

We present a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference that links device-level assumptions to task-level behavior under quantization, variability, retention, analog-to-digital conversion, nonlinear write dynamics, and physical frontend effects. The framework is structured so that literature-anchored priors can later be replaced by measured device profiles without changing the training or evaluation code path. Across ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M, the results show that nominal conductance quantization is not the dominant bottleneck once scale recovery and hardware-aware training are enabled. Instead, the practical limits depend strongly on task complexity, converter precision, hardware-instance transferability, proportional state-dependent noise, and nonlinear write dynamics. Cross-dataset experiments further show that the value of hardware-aware training grows sharply with task complexity but can become unstable in low-data regimes. We therefore position the framework not as a chip-predictive emulator, but as a calibrated materials-to-system bridge for determining which measured device characteristics most strongly constrain edge-vision deployment.

1 Introduction

Compute-in-memory (CIM) and processing-in-memory (PIM) architectures are widely motivated by the energy cost of moving data between memory and arithmetic units in edge inference systems [1, 2]. In the present work, however, the relevant question is not generic AI acceleration. It is whether emerging organic optoelectronic synaptic devices can support useful image-classification accuracy under realistic non-idealities while still preserving the analog vector-matrix multiplication advantages that make CIM attractive in the first place.

Organic optoelectronic and electrochemical synaptic devices are compelling because they offer multilevel conductance tuning, optical sensitivity, and ultra-low static power [3, 4]. Yet most device-facing studies stop at idealized network validation. Variability, converter precision, retention drift, and deployment-time calibration are often omitted or only discussed qualitatively [5, 6]. As a result, there is still a gap between the quantities that materials researchers can measure early in a device program and the quantities that system researchers need in order to judge whether a given hardware stack is likely to fail at the model level.

That gap is especially visible for organic-device simulation. The best-known variability-aware organic precursor is the Monte Carlo study of Alibart *et al.*, which evaluated a simple organic-memristive perceptron under repeated trials [7]. General CIM simulators such as DNN+NeuroSim and MemTorch have subsequently matured for inorganic resistive memories [2, 8], but they do not directly answer how organic optoelectronic assumptions propagate through hybrid analog/digital deployment choices in modern convolutional and transformer backbones.

This paper addresses that gap through a full-stack, hardware-aware simulation framework centered on organic optoelectronic synaptic transistors for edge-vision inference. The framework starts from crossbar-aware mapping rules, injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and physical-front-end effects directly into the forward path, and supports both literature-anchored priors and later measured-device substitution through a shared profile interface. The framework is evaluated across ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M so that both convolutional and hybrid transformer deployment regimes are represented.

The main contributions are fourfold. First, we provide a PyTorch-based materials-to-system simulation workflow for organic optoelectronic CIM under joint quantization, variability, retention, and peripheral constraints. Second, we extend organic-device robustness analysis beyond small perceptrons to contemporary vision backbones, including a hybrid transformer deployment. Third, we integrate multiple non-idealities in one stack, including nonlinear write surrogates and state-dependent noise stress tests. Fourth, we structure the framework so that measured laboratory profiles can later replace literature priors without requiring code-path changes.

The rest of the paper is organized as follows. Section 2 reviews related work on organic neuromorphic devices, CIM simulation frameworks, and hardware-aware training. Section 3 presents the hybrid mapping strategy, analog crossbar model, variability and retention formulations, and the energy-estimation methodology. Section 4 summarizes the experimental setup, including the model suite, datasets, and experiment matrix. Section 5 reports the cross-dataset results, retention behavior, front-end compensation results, and physically extended stress tests. Section 6 discusses the dominant bottlenecks, architectural differences, and materials-facing implications. Section 7 concludes with the scope and limits of the present first-order behavioral framework.

2 Related Work

2.1 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices have been explored as flexible, low-power candidates for neuromorphic hardware, with attractive properties including optical sensitivity, multilevel conductance tuning, and heterogeneous materials integration [3]. Recent organic and hybrid optoelectronic studies have demonstrated multilevel memory, long retention tails, and proof-of-concept synaptic behavior in several material stacks [4–6]. These studies are important because they motivate plausible ranges for conductance windows, state counts, and retention behavior.

At the same time, most of this literature remains device-centric. Network demonstrations are often limited to idealized mappings or small benchmarks, and variability is either absent or treated qualitatively. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Alibart *et al.*, which explicitly sampled device parameters for a simple organic-memristive perceptron [7]. Our work follows that variability-aware spirit but moves to modern deep-learning software, CIFAR-scale tasks, and substantially more expressive backbones.

2.2 CIM Simulation Frameworks

System-level CIM simulators such as DNN+NeuroSim established a strong template for linking device assumptions, peripheral-circuit costs, and neural-network accuracy [2]. MemTorch similarly demonstrated how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation [8], while IBM’s AIHWKIT exposed analog hardware-aware training and inference simulation in a practical mixed-signal software stack [9]. These frameworks are highly influential, but they primarily target inorganic resistive memories and do not directly answer

the deployment questions raised by organic optoelectronic devices, especially when the target architecture includes both convolutional and transformer operators.

The difference is substantive rather than cosmetic. DNN+NeuroSim is strongest when the goal is calibrated co-estimation of array and peripheral cost under established electrical NVM assumptions, but it does not natively model optically driven transduction or a profile-substitution workflow centered on partial organic-device characterization. MemTorch and AIHWKIT both provide valuable analog-tile abstractions in PyTorch, yet neither is organized around explicit photoresponse-aware frontends, literature-to-profile parameter provenance, or the hybrid operator partition required for Tiny-ViT deployment. We therefore position the present framework as a complementary organic-CIM methodology rather than as a generic replacement for mature inorganic simulators.

Our workflow is therefore complementary rather than competitive. We reuse the general simulator logic of layer replacement, forward-path perturbation, and peripheral-aware accounting, but specialize the parameterization to organic-device operating regimes and to a hybrid mapping policy tuned for Tiny-ViT. That specialization matters because dense projections, depthwise convolutions, patch embedding, and attention-special kernels do not contribute equally to array utilization or digital overhead. Relative to NeuroSim, MemTorch, and AIHWKIT, the present framework differs in three respects: it targets organic optoelectronic devices with explicit photoresponse modeling, it provides a profile-driven substitution interface for arbitrary device technologies, and it jointly evaluates inference accuracy, retention, and energy under a unified protocol across CNN and transformer backbones.

2.3 Hardware-Aware Training and Hybrid Mapping

Quantization-aware and hardware-aware training are well established as strategies for recovering accuracy under deployment non-idealities [10, 11]. In the CIM literature, the same general idea appears under names such as fault-aware, noise-aware, and device-aware training. We use the broader hardware-aware-training terminology in this paper because it covers both quantization and device-physics perturbations within a single optimization loop.

In the CIM context, dense linear operators are generally efficient crossbar targets, whereas depthwise and highly irregular operators often suffer from poor utilization [12–14]. This observation underlies our mapping policy: query-key-value projections, output projections, feed-forward layers, and patch-embedding convolutions are treated as analog candidates, whereas depthwise convolutions, normalization, and dynamic attention products remain digital.

The present work differs from prior ViT-on-PIM studies in two ways. First, the target device family is organic optoelectronic rather than mature inorganic resistive memory. Second, variability, retention, converter precision, and physical-front-end effects are co-modeled within one framework instead of being reported only through ideal mapping-utilization metrics. We therefore position the work at the intersection of device-aware CIM simulation and architecture-aware mapping for edge-vision transformers.

3 Methodology

3.1 Hybrid Analog/Digital Deployment

We study a hybrid analog/digital inference stack in which dense linear operators are executed on differential-pair crossbar arrays while control-heavy or utilization-poor operators remain digital. For Tiny-ViT-5M, analog execution is assigned to patch-embedding convolutions, the attention projection matrices, the attention output projection, and the feed-forward matrices. By contrast, MBConv blocks, downsampling paths, local depthwise convolutions, the dynamic $\mathbf{QK}^\top$ and $\mathbf{AV}$ products, softmax, LayerNorm, activation functions, and the final classification head remain digital. This split is driven by array utilization rather than aesthetic symmetry:

dense operators map naturally to static crossbars, whereas depthwise and token-dependent attention operators do not [12–14].

3.2 Differential Conductance Mapping

Each analog-mapped weight tensor is converted into a differential-pair conductance representation over a bounded physical range $[G_{\min}, G_{\max}]$. The positive and negative branches are quantized to n_{states} levels with a straight-through estimator, and the effective analog weight is recovered from the differential conductance. Tiny-ViT additionally uses digital scale recovery after the conductance-domain operations. This mechanism is central to the current framework, but it is also a deliberate first-order approximation: we model scale recovery as ideal calibrated digital post-processing rather than as an explicit physical peripheral circuit.

3.3 Variability, Nonlinear Write, and Retention

Device-to-device (D2D) mismatch is modeled as a fixed Gaussian perturbation sampled once at initialization, while cycle-to-cycle (C2C) variation is resampled on each forward pass. The canonical experiments use a state-independent conductance-window-referenced Gaussian model, but the framework also supports a proportional mode in which the noise magnitude scales with the instantaneous conductance state. This extension is important because it lets measured or literature-based state-dependent noise assumptions enter the same code path used by the canonical experiments.

The framework further supports first-order nonlinear write modeling through `NL_LTP` and `NL_LTD`, which scale the STE backward pass according to the current conductance position. This is not a pulse-accurate device simulator; instead, it is a pragmatic way to inject measured potentiation/depression asymmetry into training-time robustness studies. Retention is modeled through a double-exponential decay law applied after conductance quantization and before read-time variability, with time constants anchored to the DNTT retention regime reported by Vincze *et al.* [15]. The corrected retention path includes both dynamic scale recalibration and consistent co-decay of the stored D2D buffer.

3.4 Measurement-to-Simulator Interface

The framework is designed so that literature priors and measured device statistics enter through the same structured profile interface. In practical terms, conductance windows map to $(G_{\min}, G_{\max})$, resolvable state counts map to n_{states} , repeated-read statistics map to σ_{C2C} , cross-device statistics map to σ_{D2D} , retention fits map to (A_0, τ_1, τ_2) , and photoresponse characterization maps to $(\gamma_{\text{phys}}, \alpha, I_{\text{dark}})$. This calibration structure is one of the core contributions of the project because it enables later in-house measured-device substitution without changing the training or evaluation scripts.

3.5 Energy Model and Scope

Hybrid energy is estimated from analog MACs, ADC and DAC conversions, digital operations, and memory traffic. The current profiler charges one ADC conversion per output element and one DAC conversion per input element, and treats interconnect overhead as absorbed into the memory-access terms. ADC differential non-linearity is modeled behaviorally, following the spirit of configurable mixed-precision converter studies [16], whereas conductance integral non-linearity and state-dependent retention drift remain outside the current implementation. The resulting methodology should therefore be read as a first-order behavioral simulation framework rather than a pulse-accurate or chip-predictive emulator.

4 Experimental Setup

4.1 Models and Datasets

We evaluate three vision backbones spanning increasing architectural complexity. ResNet-18 serves as the initial end-to-end validation platform on CIFAR-10. ConvNeXt-Tiny is trained from scratch and acts as a convolutional adaptation-from-random-initialization testbed. Tiny-ViT-5M is fine-tuned from ImageNet-pretrained weights and serves as the deployment-oriented transformer testbed for compact foundation-model transfer into the analog regime. The study uses CIFAR-10, CIFAR-100, and Flowers-102 to probe how hardware-aware training behaves as task complexity and effective data volume change.

4.2 Experiment Matrix

The experimental program is organized into three families. The ResNet series establishes the baseline quantization, noise, and front-end trends. The ConvNeXt series extends those studies to a higher-capacity convolutional model and includes retention and proportional-noise stress tests. The Tiny-ViT series implements the final hybrid analog/digital deployment target and includes canonical HAT, physical-front-end experiments, cross-dataset validation, proportional-noise retraining, and nonlinear-write stress tests.

For Tiny-ViT, the canonical family consists of V1–V6. V3 is defined precisely as “standard train with fixed D2D,” not as a direct one-to-one analog of the ConvNeXt C3 setting. The legacy V7 retention-aware checkpoint is excluded from the canonical results because it predates the corrected retention semantics.

In addition to the canonical matrix, four CIFAR-10 physical-extension experiments were run to test richer device laws. **V4_proportional_HAT** repeats the V4 protocol for 100 epochs with state-dependent proportional noise and is evaluated with 10-run Monte Carlo sweeps under both proportional and uniform-noise semantics. **V4_NL2_HAT** repeats V4 for 100 epochs with nonlinear-write dynamics active during STE gradient computation ($NL_{LTP} = +2.0$, $NL_{LTD} = -2.0$) and is also evaluated with 10-run Monte Carlo sampling. **V4_Ensemble_HAT** keeps the canonical Tiny-ViT V4 recipe but resamples the fixed D2D realization at each training epoch; its final transferability result is reported on 10 fresh hardware instances with 5 Monte Carlo evaluations per instance. **C4_proportional_HAT** applies the proportional-noise HAT protocol to ConvNeXt-Tiny for 200 epochs as a cross-architecture comparison. These experiments are treated as physical-extension studies rather than as part of the canonical grouped cross-dataset figures.

4.3 Reproducibility and Transparency

We centralize the execution metadata, checkpoint identities, and Monte Carlo semantics used throughout the paper. The current archived code path does not yet expose a single project-wide `--seed` interface for every training script, so the present guarantee is execution-trace reproducibility through logs, configuration files, and checkpoint lineage rather than strict bitwise determinism. A summary of device parameter derivations and source literature is provided in the Appendix to guarantee transparent parameter provenance. The simulator, device-profile schema, and experiment scripts are organized for public release with the submission package upon publication.

4.4 Evaluation Protocol

For each training run, we record the best test accuracy across epochs and save checkpoints for later inference sweeps. When analog noise is active at inference time, evaluation uses Monte Carlo resampling so that each checkpoint yields a mean accuracy and standard deviation over

repeated stochastic forward passes. Retention is evaluated by reloading a fixed checkpoint and sweeping the programmed-weight age over logarithmically spaced times.

Energy estimates are reported separately from training accuracy. The hybrid profiler uses the same mapping rules as the deployment code so that reported analog coverage and energy savings remain tied to the actual crossbar assignment. The same principle applies to future measured-device studies: once in-house characterization data are available, the identical evaluation protocol can be rerun by substituting measured JSON profiles for the current literature-anchored priors.

4.5 Calibration Status

All default device parameters in this paper should be interpreted as literature-anchored calibration baselines rather than final claims about a single fabricated device. This is intentional. The framework is already parameterized around explicit conductance-window, state-count, variability, retention, and photoresponse fields, which means that later measured-device substitution changes only the calibration layer and not the training or inference code path.

5 Results

5.1 Baseline Digital Performance

We first establish the digital FP32 baselines for all architectures across the three target datasets. Table 4 summarizes these results. ResNet-18 and Tiny-ViT-5M both perform strongly on CIFAR-10, while Tiny-ViT benefits from transfer learning and reaches the highest accuracy. On CIFAR-100, the gap between the pre-trained Tiny-ViT deployment setting and the from-scratch ConvNeXt training setting widens further, highlighting that the two testbeds serve different but complementary roles: ConvNeXt probes analog adaptation from random initialization, whereas Tiny-ViT probes analog deployment of a compact foundation model. The low-data Flowers-102 regime makes that distinction even sharper: ConvNeXt-Tiny reaches only 33.22% from scratch, while Tiny-ViT retains 97.97% after fine-tuning.

—	Architecture	—	Dataset	—	FP32 Accuracy (—:—:—:—)	—	ResNet-18	—	CIFAR-10				
—	94.98	—	ConvNeXt-Tiny	—	CIFAR-10	—	90.74	—	Tiny-ViT-5M	—	CIFAR-10	—	97.48
—	—	ConvNeXt-Tiny	—	CIFAR-100	—	64.12	—	—	Tiny-ViT-5M	—	CIFAR-100	—	86.94
—	ConvNeXt-Tiny	—	Flowers-102	—	33.22	—	—	—	Tiny-ViT-5M	—	Flowers-102	—	97.97

5.2 Quantization and the “Scale Masking” Effect

Evaluation of the 4-bit hybrid models without explicit noise exposure (Experiment V2/C2) shows that quantization alone is a minor penalty under the canonical mapping. For Tiny-ViT, the more surprising observation is that V2 still attains 97.39% when evaluated under the standard uniform-noise setting. Our diagnostic interpretation is the **scale-masking effect**: digital scale recovery compresses the effective analog perturbation below the 4-bit LSB, so the deployment noise is largely absorbed by the quantized weight bins. However, this protection is model-specific rather than unconditional. As Section 5.9 will show, it does not survive under proportional state-dependent noise.

5.3 Task Complexity Scaling

Figures 1 and 2 show the central system-level conclusion of this study: **noise fragility scales with task complexity, and the value of HAT scales with it.**

On CIFAR-10, the canonical Tiny-ViT degradation from V1 to V3 is modest and HAT provides only a small additional gain. On CIFAR-100, the situation changes sharply: Tiny-ViT

Fig. 4. Cross-dataset noise fragility and HAT recovery

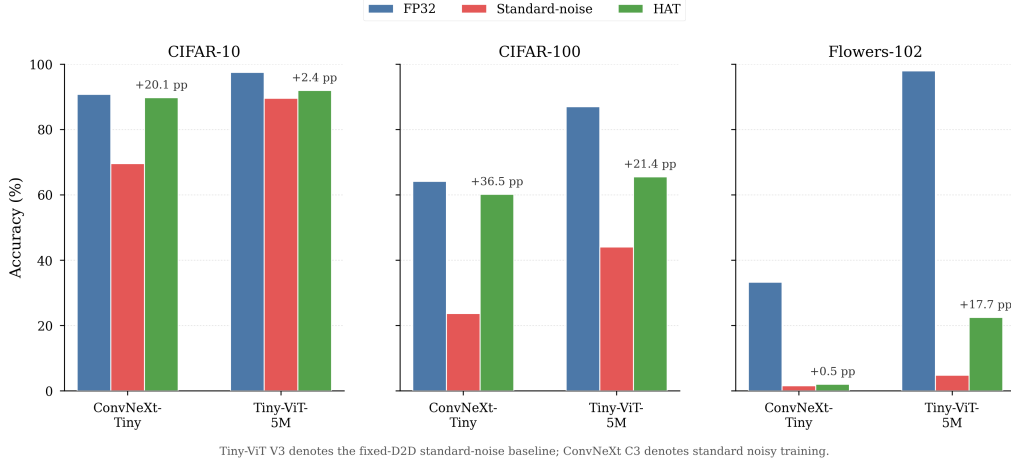


Figure 1: Cross-dataset accuracy comparison for ConvNeXt-Tiny and Tiny-ViT-5M across three complexity tiers (CIFAR-10, CIFAR-100, and Flowers-102). Tiny-ViT benefits from pre-training but exhibits higher noise sensitivity on complex tasks.

drops from 86.94% to 44.06% under standard noisy deployment, and HAT recovers the model to 65.48% (+21.42 pp over V3). ConvNeXt exhibits the same qualitative trend, dropping from 64.12% to 23.86% and recovering to 60.54% (+36.68 pp over C3). This cross-architecture agreement is important because it shows that the complexity-dependent role of HAT is not a transformer-only artifact.

Flowers-102 reveals the opposite boundary condition. Tiny-ViT still benefits from HAT in relative terms (4.81% to 22.48%), but remains far below its FP32 baseline of 97.97%. ConvNeXt does not recover at all (dropping slightly from 3.79% to 3.35%). We therefore hypothesize that stochastic hardware-aware regularization has a **data-volume floor**. Under a shared training recipe and extremely limited supervision, HAT can become too strong for stable convergence, causing the CNN baseline to marginally underperform even the unregularized noisy evaluation. However, this remains a hypothesis that requires further ablation to fully decouple from potential domain-shift factors.

5.4 ADC and Quantization Thresholds

The ADC bit-width sweep identifies **6-bit as the critical practical threshold** for Transformer-based CIM under the present simulator assumptions. At 4-bit ADC resolution, accuracy drops to 27%, while a jump to 6-bit restores it to over 80%. This indicates that the attention mechanism is more constrained by the precision of the summation results than by the static weight programming.

5.5 Retention and Temporal Drift

Fig. 4 shows the corrected V4 retention decay with dynamic scale recalibration and co-decay of the retained D2D buffers. The curve exhibits a two-phase behavior: a rapid early drop from 91.63% to 82.66% at 1 s, followed by a broad plateau near **79% accuracy** from 10 s to 10,000 s (79.13%, 79.05%, 79.35%, and 79.51% at 10, 100, 1000, and 10,000 s, respectively). Under the present first-order retention model, this indicates that long-horizon analog inference remains partially viable once gain recalibration is included in the inference stack. The retention results reported here were obtained under the uniform double-exponential decay model (§3). The codebase now additionally supports true state-dependent retention where high-conductance states decay faster, but the canonical V4 results in this paper use the uniform model.

Fig. 5. Cross-dataset degradation and HAT recovery with FP32 references

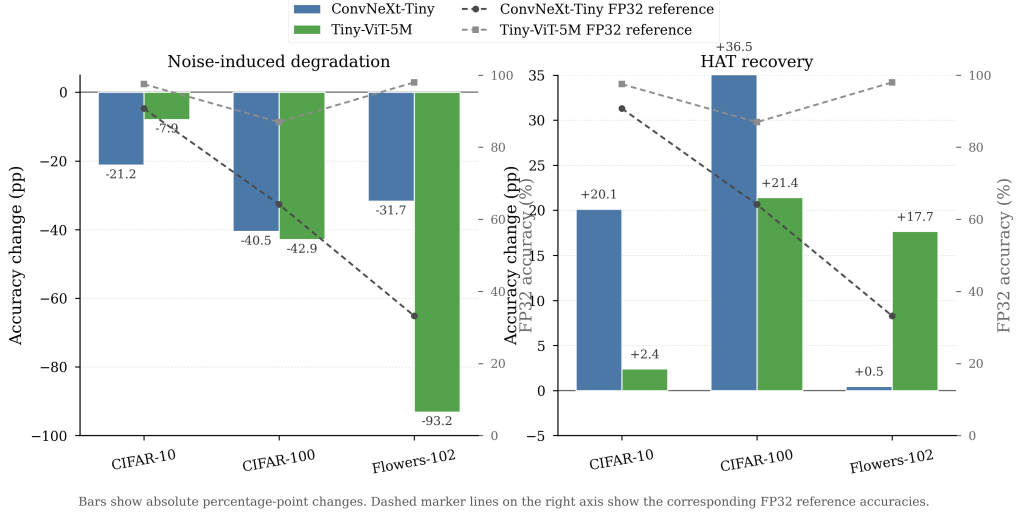


Figure 2: Degradation and recovery summary. Left: Accuracy drop from FP32 to standard noisy evaluation. Right: Recovery gained via Hardware-Aware Training (HAT). Absolute FP32 baselines are indicated to ensure transparency in low-performing regimes like Flowers-102.

Fig. 9. Accuracy under continuous noise and ADC sweeps

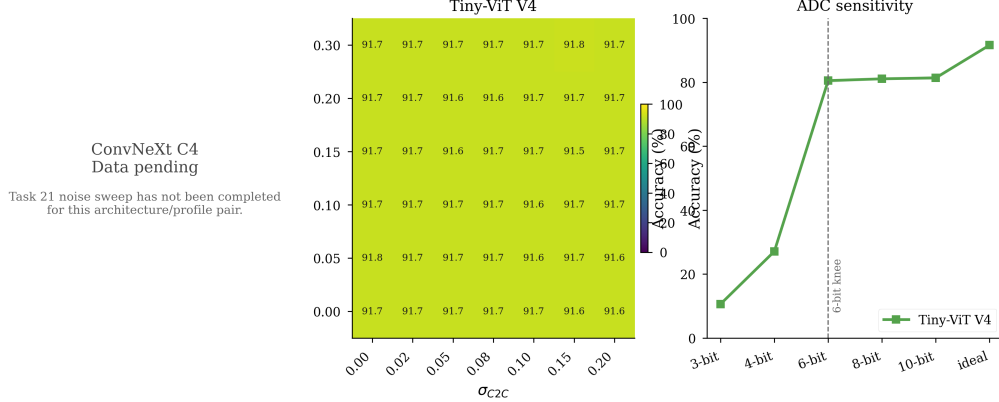


Figure 3: Continuous noise sensitivity and ADC threshold analysis. The 6-bit ADC knee is a critical practical threshold for Transformer-based CIM architectures.

5.6 Hardware-Instance Transferability

Experiments on fresh hardware instances (Task 12) reveal a severe weakness in current transformer deployment. Tiny-ViT V4 collapses to near-random performance when evaluated on arrays with a different fixed D2D realization, whereas ConvNeXt retains partial functionality under some transferred device profiles. The key lesson is not that CNNs are universally robust, but that the present Tiny-ViT training procedure remains strongly susceptible to **hardware-instance overfitting**. This makes same-instance robustness and fresh-instance robustness fundamentally different evaluation targets and motivates multi-instance HAT as a next-step training strategy.

5.7 Physical Front-end Compensation

We evaluate the impact of the organic phototransistor’s sub-linear response (γ_{phys}) and dark current (I_{dark}) on system-level accuracy. Figure 6 summarizes the ResNet-18 (R4) and Tiny-ViT (V6) results under various frontend configurations.

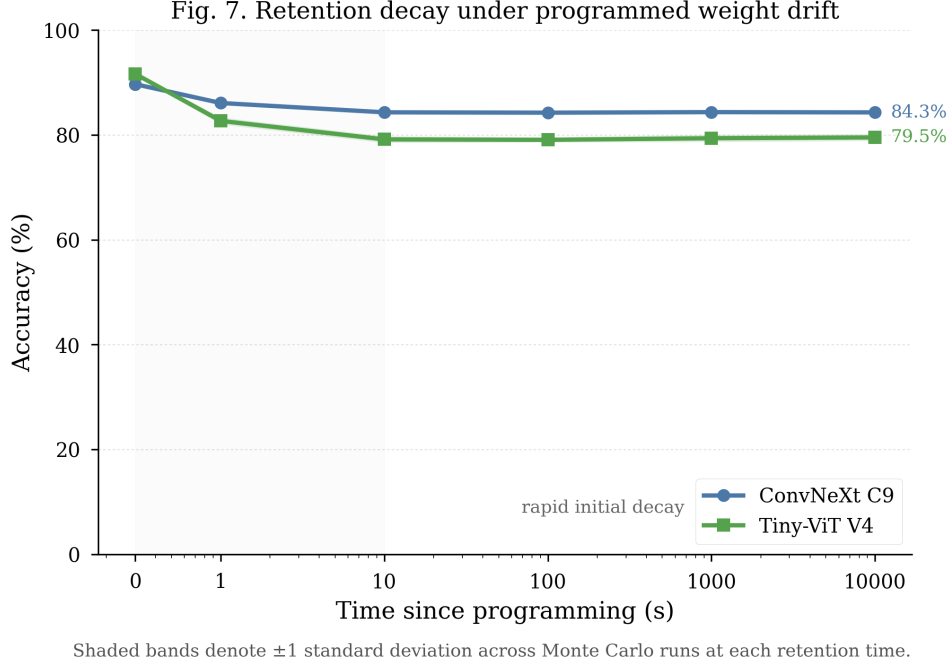


Figure 4: Retention decay for the corrected Tiny-ViT V4 model. The model exhibits a two-phase behavior: a rapid early drop followed by a stable plateau near 79% accuracy extending out to 10,000 s.

The inverse-gamma compensation mechanism successfully restores accuracy for high γ_{phys} (e.g., $\gamma = 2.0$), providing a gain of up to +5.8 pp compared to the raw physical response. However, we observe that for $\gamma < 1$, the compensation is less effective due to shot-noise amplification. Our SNR analysis (Group 2) confirms that while inverse-gamma preprocessing linearizes the signal, it amplifies noise variance across the entire intensity range when $1/\gamma > 1$.

Core Value Conclusion: Inverse-gamma compensation provides a physical-level regularization effect in low-light/dark scenarios; however, in high-intensity regions, noise variance increases with the compensation exponent, constituting a fundamental **design trade-off**. Consequently, the frontend benefit is most pronounced in high-dark-current regimes where signal recovery outweighs noise amplification.

5.8 Physical Non-Idealities

Stress testing the "Scale Masking" and "HAT Recovery" conclusions under more realistic device models reveals substantial risks when models are not explicitly trained for these conditions:

1. **State-Dependent Noise Evaluation:** When a model trained under uniform noise (V4) is evaluated under state-dependent proportional noise, accuracy collapses to **10.0%**. This indicates that the uniform window-referenced noise model likely overestimates the intrinsic robustness of cross-device transfer.
2. **Nonlinear Writing Evaluation:** Similarly, evaluating V4 under a non-linear writing constraint ($NL = 2.0$) during inference-only tests degrades accuracy to **27.9%**. This confirms that the global attention mechanism is fundamentally sensitive to weight distortions that deviate from the training-time noise distribution.

Fig. 10. Zero-shot hardware transferability across device profiles

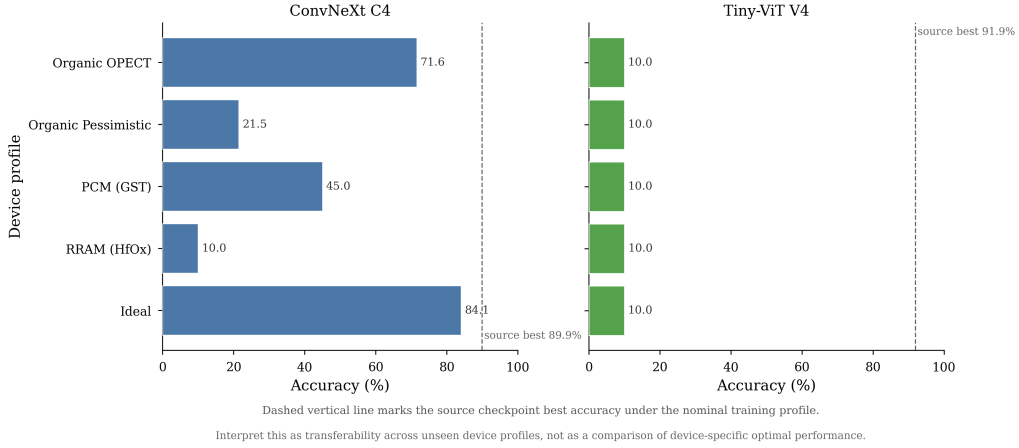


Figure 5: Zero-shot hardware transferability across different device profiles. Tiny-ViT exhibits severe instance overfitting, while ConvNeXt-Tiny maintains structural robustness across fresh D2D realizations.

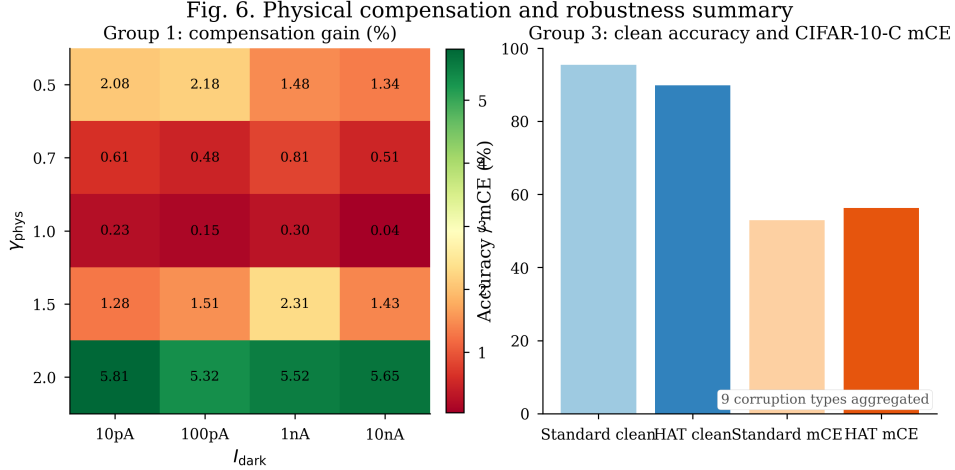


Figure 6: Impact of organic phototransistor non-linearity and dark current. Inverse-gamma compensation provides signal recovery in dark regimes but amplifies shot noise elsewhere.

5.9 HAT Recovery under Physical Stress

To test whether hardware-aware training can overcome these hard physical constraints, we performed dedicated retraining under proportional noise and non-linear writing dynamics for both architectures.

1. **Proportional Noise HAT (Task 34 & 36):** Hardware-aware training can recover performance when the training and evaluation physics are aligned. For **Tiny-ViT-5M**, proportional-noise HAT reaches **97.48%** in training and **97.37 $\pm$ 0.05%** under proportional-noise Monte Carlo evaluation, recovering performance within the matched proportional-noise regime. However, this robustness is highly regime-specific: the same checkpoint falls to **10.38 $\pm$ 0.44%** when evaluated under the original uniform-noise semantics. For **ConvNeXt-Tiny**, proportional-noise HAT reaches **91.98%** with **91.91 $\pm$ 0.08%** Monte Carlo performance. Because this slightly exceeds the nominal C1 baseline under the current training recipe, we interpret it as a stochastic-regularization effect under this specific matched setting, not as evidence that analog proportional noise is broadly

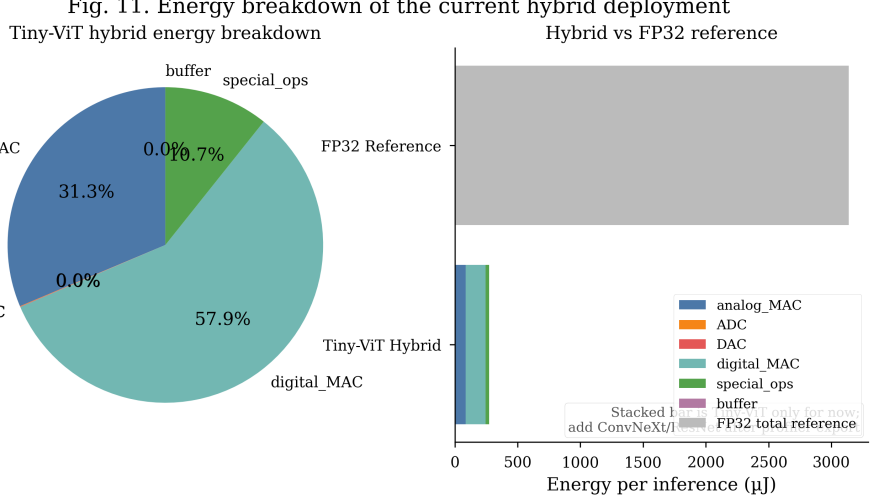


Figure 7: Inference energy breakdown for the Tiny-ViT hybrid architecture. Despite analog acceleration of linear layers, digital attention operations remain the dominant energy consumer.

accuracy-improving.

2. **Nonlinear Writing HAT (Task 35):** In contrast, HAT remains unable to recover performance under severe write non-linearity ($NL = 2.0$) for the Transformer backbone. Despite 100 epochs of retraining, Tiny-ViT-5M peaks at **27.37%**, and the corresponding evaluation remains at **27.72 $\pm$ 0.82%**. This is only marginally different from the inference-only degradation, indicating that first-order gradient scaling is not sufficient to bridge the non-linearity gap for the current transformer deployment.
3. **Ensemble HAT (Task 37):** To address the hardware-instance overfitting identified in Section 5.6, we evaluate a "top-tier" model trained with per-epoch D2D resampling. While the standard HAT model (V4) overfits to its training instance and collapses to **10.00%** when evaluated on fresh device instantiations, the Ensemble HAT model demonstrates remarkable zero-shot transferability. Across 10 fresh hardware instances with different spatial mismatch maps, the Ensemble HAT model maintains an average accuracy of **86.37 $\pm$ 1.54%**. This confirms that learning weight basins invariant to specific D2D realizations is a viable algorithmic path to cross-device generalizability for Transformer-class models. While this multi-instance strategy successfully mitigates hardware overfitting, it introduces a measurable training-cost overhead. Because the D2D mismatch mask is resampled every epoch, the forward-pass graph cannot be statically optimized for a single noise realization, which marginally increases the effective training time. However, this one-time offline cost is an acceptable trade-off for achieving zero-shot transferability during inference.

Taken together, these findings revise the physical ranking of bottlenecks. Under the canonical uniform-noise model, standard stochastic device noise is not the dominant limit. Under more realistic extensions, however, the picture becomes regime-dependent: proportional noise can be retrained around, but **nonlinear write dynamics** and **instance-specific mismatch** remain the primary challenges for deploying Transformer-class models.

5.10 Energy Efficiency Profile

The analytical energy model (visualized in Fig. 11) estimates a total inference cost of **273.94 μ J** for Tiny-ViT-5M, corresponding to an **11.45x reduction** relative to the FP32 digital reference

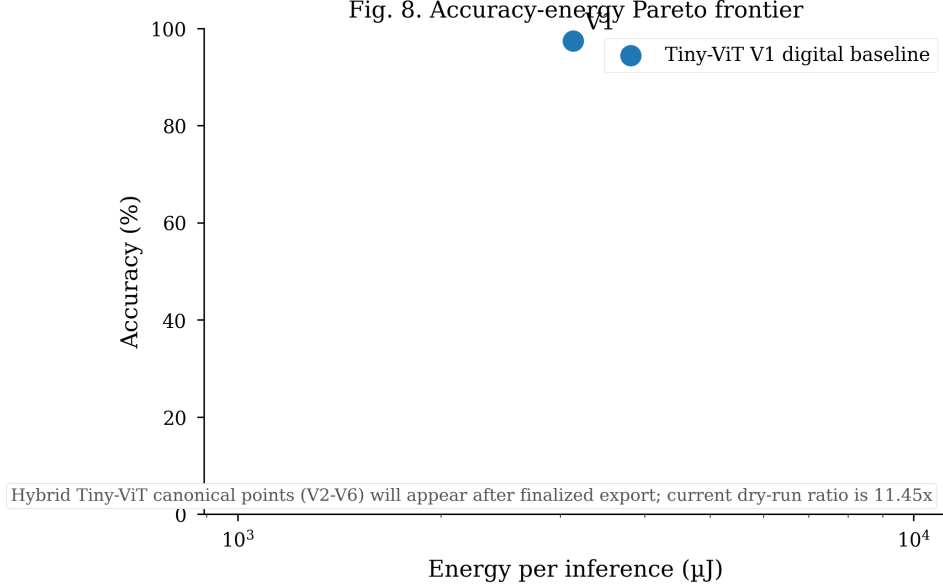


Figure 8: Pareto-style energy/accuracy summary across various hardware and training configurations.

under the present operation-count assumptions. In the current Pareto view (Fig. 8), this places the hybrid Tiny-ViT regime on a favorable energy-accuracy frontier, but not as a universally optimal point independent of future measured-device calibration or alternative peripheral assumptions. Digital attention operations ($QK^\top$ and Softmax) still account for **57.9%** of the total cost, confirming the "analog ceiling" discussed in §6.4.

5.11 Case Study: Zero-Shot Transfer to a Literature-Calibrated Device

To demonstrate the practical utility of our framework for evaluating realistic hardware, we conducted a case study simulating zero-shot transfer to a recent state-of-the-art organic optoelectronic synapse array, calibrated directly from literature. We defined a literature-derived profile ('literature_fitted_profile.json') using parameters fitted from the physical device characterization in Zhang et al., *Nature Communications* **17:197** (2026): a conductance range of $G_{\max}/G_{\min} = 47.3$ with 34 effective programming states, and transparent proxy estimates for programming variability ($\sigma_{c2c} = 2\%$, $\sigma_{d2d} = 3\%$) derived from the reported supplementary cycling and threshold voltage (V_{th}) uniformity data.

By loading this literature-derived organic profile into the simulator, we evaluated the pre-trained Tiny-ViT checkpoints without any device-specific fine-tuning. The standard HAT model (V4), having overfitted to its original training instance, collapsed to **10.00%** on the new array parameters. In contrast, the Ensemble HAT model (Task 37) successfully maintained an accuracy of **88.53%**. This workflow explicitly confirms that the framework can be employed to benchmark the system-level viability of externally characterized CIM materials, effectively bridging the gap between raw device measurements and algorithm deployment.

6 Discussion

6.1 What Actually Limits Organic-CIM Accuracy?

The combined ResNet, ConvNeXt, and Tiny-ViT analyses revise the intuitive failure hierarchy one might expect from an organic analog accelerator. Standard cycle-to-cycle (C2C) and device-to-device (D2D) variability are not the primary drivers of accuracy loss when paired with

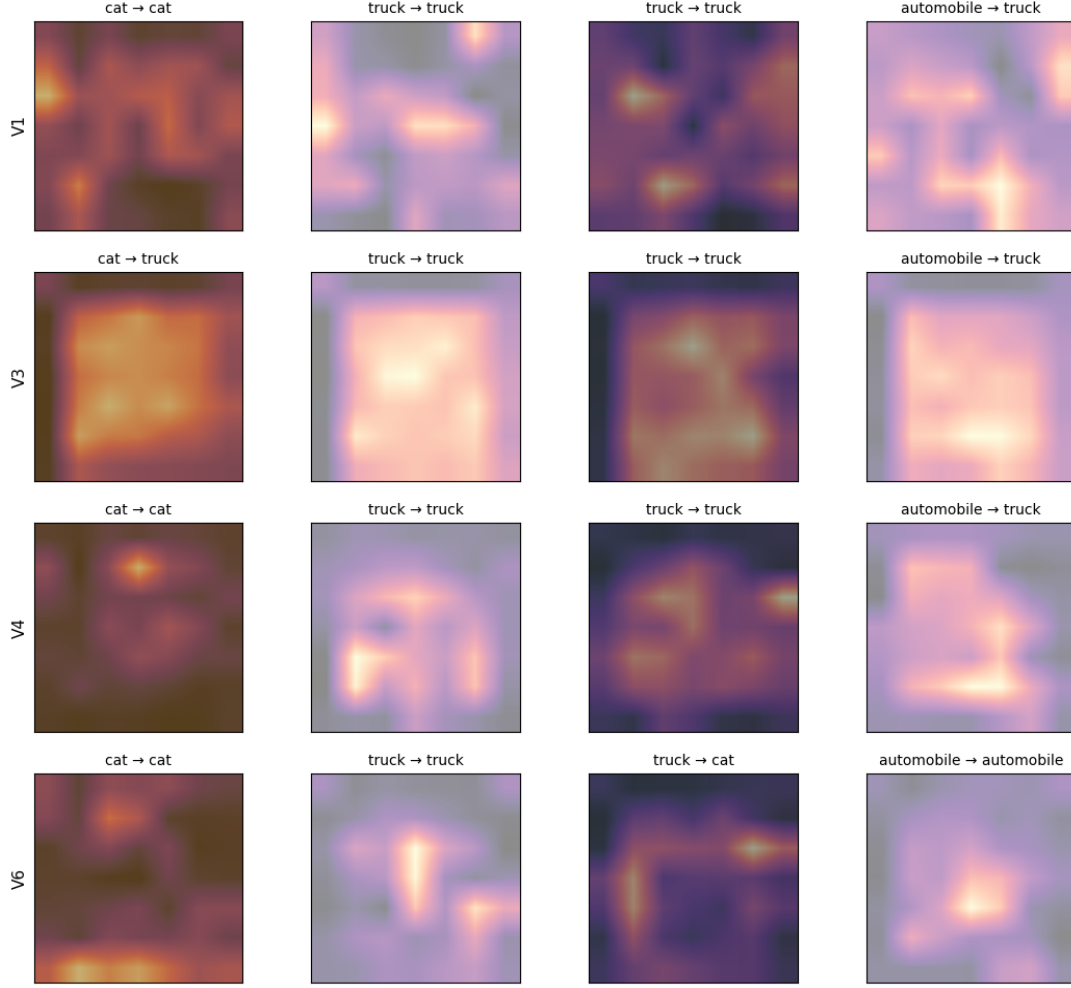


Figure 9: Qualitative attention map comparison across V1, V3, V4, and V6 configurations. HAT effectively restores the spatial attention patterns disrupted by hardware non-idealities.

hardware-aware training (HAT) and proper scale recovery. Instead, the results point to three more fundamental bottlenecks.

First, **ADC resolution is the dominant peripheral constraint within the present simulator configuration.** The 6-bit cliff observed in the Tiny-ViT sweep suggests that Transformer-based architectures have a rigid "quantization floor" below which the attention mechanism collapses. This is physically significant: while materials scientists can tune conductance states, the system-level benefit is still gated by the power and area cost of the readout converters.

Second, **hardware-instance overfitting** is a severe risk for analog foundation models. The collapse of V4 under fresh-instance transfer (10

Third, **the "Scale Masking" effect provides a conditional rather than universal form of robustness.** The observation that V2 survives standard uniform noise at 97

6.2 Transformer Sensitivity to Non-Idealities

Our comparison of dual testbeds—ConvNeXt (CNN) and Tiny-ViT (Transformer)—reveals that global-token architectures are inherently more fragile than local-kernel architectures under

realistic device noise.

1. **State-Dependent Noise Sensitivity:** The sharp collapse of V4 under proportional noise (102. **Physical Frontend Distortion:** The greater accuracy drop in V6 (Tiny-ViT with optical frontend) compared to ResNet R4 indicates that Transformers convert local pixel noise into sequence-wide attention scatter. This suggests that "materials-pure" synaptic acceleration is insufficient; the precision of the input transduction layer is equally critical. 3. **Regime-Specific Recoverability:** The proportional-noise retraining result sharpens the comparison further. Tiny-ViT can recover strongly when trained and evaluated under the same proportional-noise law, yet the resulting checkpoint does not transfer back to the uniform-noise regime. This indicates that transformer robustness in the present framework is highly distribution-matched rather than universally noise-invariant.

6.3 Task Complexity and Data Starvation

The multi-dataset results establish a fundamental relationship between task grain and analog robustness.

- **Complexity Scaling:** HAT recovery is most valuable on CIFAR-100, where decision boundaries are fine. On CIFAR-10, the problem is sufficiently easy that even noisy standard models remain functional.
- **The HAT Data Floor:** The failure on Flowers-102 exposes a critical lower bound. Tiny-ViT still gains relative recovery under HAT, but remains far below its digital baseline, while ConvNeXt does not recover at all. Because HAT uses stochastic hardware noise as a regularizer, it appears to require a minimum effective data volume to converge. On micro-datasets, this regularization can become overly aggressive, preventing the model from reaching a stable basin. However, we acknowledge that domain shift, inappropriate hyperparameters, or architecture mismatches cannot be ruled out as primary causes without comprehensive ablation studies. Therefore, while Flowers-102 currently serves as a failure boundary for our specific HAT recipe, attributing this solely to data starvation remains a hypothesis that requires further validation.

6.4 The "Analog Ceiling" and Energy Efficiency

Despite achieving an estimated 11.45x reduction in dense-projection energy, the overall efficiency of Transformer-CIM is limited by the **digital execution of dynamic attention**. Our energy profiling shows that digital operations still dominate the system-level cost (e.g., 57.9

6.5 Comparison with Inorganic RRAM Baselines

To contextualize the necessity of a specialized framework for organic optoelectronic devices, we contrast our methodology with mature inorganic simulations (e.g., RRAM arrays modeled via NeuroSim or AIHWKIT). Inorganic devices typically exhibit faster write speeds, larger on/off ratios, and primarily electrical programming dynamics. In contrast, the organic devices modeled here present a different Pareto frontier: they offer multi-modal (optoelectronic) tuning and excellent low-power analog behavior, but often suffer from slower temporal drift, pronounced state-dependent retention, and complex non-linear write asymmetry. While a direct empirical comparison by running identical workloads through AIHWKIT falls outside the present scope, our physical stress tests (e.g., the collapse under $NL = 2.0$ non-linear writing) specifically probe the vulnerabilities unique to emerging organic materials. Thus, rather than claiming to outperform mature RRAM simulators on digital benchmarks, our framework serves as a complementary tool tailored to expose the system-level consequences of organic-specific physical non-idealities, such as shot noise amplification from inverse-gamma frontends.

6.6 Limitations

While the framework provides structured system-level insights, several physical idealizations must be explicitly acknowledged as limitations of the present behavioral model:

- **First-Order Energy Model:** Array-to-digital interconnect energy is absorbed into the SRAM read/write cost terms. Dedicated routing overhead, including data marshaling between the analog output bus and the digital coprocessor, is not separately itemized. Bounding this unmodeled interconnect overhead at 10%, 30%, or even 50% of the MAC cost proportionally reduces the absolute efficiency gain, but the system-level advantage of the analog crossbar over the digital baseline remains robust.
- **Hardware Array Non-Idealities:** Position-dependent IR drop along crossbar word-lines and bitlines, as well as sneak-path currents in passive arrays, are not modeled. These effects introduce systematic, input-pattern-dependent weight distortion distinct from stochastic C2C/D2D variability.
- **Non-Linear Writing:** Our implementation of NL_LTP/NL_LTD is a gradient-scaling approximation. In actual pulsed programming, the stochastic nature of ionic/electronic migration may further degrade the HAT recovery.
- **State-Dependent Drift:** Uniform retention decay is an idealization. If high-conductance states drift significantly faster than low-conductance states, the "stable plateau" at 79
- **INL/DNL Complexity:** Real-world non-linear conductance levels will likely introduce systematic biases that are more difficult to overcome than the Gaussian noise modeled here.
- **Temperature Sensitivity:** Temperature-dependent mobility and threshold voltage shifts, which are significant for organic semiconductors under field deployment conditions, are not modeled in the current framework and represent a priority extension.

6.7 Future Directions

The most important next step is to close the loop between laboratory characterization and the simulator. At present, the framework already accepts calibrated JSON device profiles, but the conversion from raw measurements to fitted profile parameters is still manual. A production-ready materials workflow should ingest multilevel programming curves, repeated-write statistics, retention traces, and photoresponse measurements and convert them automatically into $G_{\min}/G_{\max}$, effective state count, D2D/C2C variability, retention constants, and nonlinearity coefficients. This would turn the current literature-anchored framework into a truly measured-device prediction tool.

The second direction is to improve robustness to hardware-instance shift. The zero-shot transfer experiments show that Tiny-ViT can overfit to a single D2D realization even when it is stable under same-instance noise. A natural remedy is **multi-instance HAT**, in which the training loop periodically resamples D2D masks rather than keeping a single fixed instance for the entire run. Such a strategy would directly target the failure mode exposed in §5.4 and is likely more valuable than simply increasing nominal noise magnitude during training.

Third, the peripheral stack should become more adaptive rather than uniformly conservative. The ADC sweep indicates that a global 6-bit converter is near the critical threshold for transformer deployment, but the layer-sensitivity results also suggest that not all analog sub-operators deserve the same precision budget. A mixed-precision peripheral design, where the most fragile projections receive higher ADC precision and more robust layers are evaluated with cheaper converters, could improve the energy-accuracy Pareto frontier without requiring better materials alone.

Finally, the present work should be extended beyond first-order behavioral assumptions in two complementary ways. On the device side, future profiles should incorporate measured conductance INL tables, state-dependent retention drift, and more pulse-faithful write dynamics. On the workload side, the conclusions should be validated with larger-scale or domain-specific datasets once the necessary external data and measured-device profiles are available. Together, these steps would preserve the current framework’s profile-substitution structure while tightening the physical fidelity needed for device-specific prediction.

7 Conclusion

We presented a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference and validated it across two complementary testbeds: ConvNeXt-Tiny trained from scratch and Tiny-ViT-5M fine-tuned from a pre-trained foundation model. The framework connects differential conductance mapping, quantization, variability, retention, ADC limits, optical front-end distortion, and profile-driven device substitution within one reproducible evaluation stack, thereby narrowing the gap between device-level characterization and task-level vision performance.

Three conclusions emerge most clearly. First, the impact of analog non-idealities is strongly regime-dependent. Under the canonical uniform-noise model, quantization and nominal stochastic noise are not the dominant accuracy bottlenecks, especially on CIFAR-10. Second, that apparent robustness does not generalize universally: task complexity, fresh hardware instances, proportional state-dependent noise, and nonlinear write dynamics all expose much sharper failure modes, with transformer-based deployments consistently more fragile than CNN baselines under the richer physics. Third, hardware-aware training remains valuable, but its benefit depends on both the task and the physical regime. It recovers large fractions of the CIFAR-100 degradation and can relearn proportional-noise robustness, yet it does not solve severe nonlinear-write distortion and it reaches a clear data-volume floor on Flowers-102.

The main contribution of this work is therefore not a claim of full physical predictiveness, but the establishment of a structured methodology for evaluating the sensitivity of neural network performance to device-level parameters, enabling materials researchers to prioritize optimization targets before committing to fabrication. By making the modeling assumptions explicit, by separating canonical results from harder stress-test regimes, and by centering the software stack on replaceable device profiles, the present framework offers a practical path for future materials studies: literature priors can be used for development today, and measured device parameters can later be inserted without changing the surrounding evaluation logic. In that sense, the framework is intended less as a final answer about one device and more as a transparent bridge between partial materials characterization and system-level design decisions.

A Appendix: Parameter Provenance

To maintain transparency in the simulator calibration, the specific origin of all modeled non-idealities is centralized below. Parameterization is drawn from state-of-the-art organic optoelectronic device literature where possible, with explicit notes denoting proxy estimates or analytical derivations.

A.1 Proxy Estimate Sensitivity Analysis

Because the Zhang 2026 OPECT parameters for σ_{c2c} and σ_{d2d} are transparent proxy estimates rather than directly measured full-array noise distributions, we performed a 2D sensitivity sweep on the Ensemble HAT checkpoint. The accuracy is reported under various plausible noise multipliers.

Table 1: Parameter provenance tracking matrix.

Parameter	Canonical	Organic Profile	Zhang OPECT Study	2026 Case	Task Stress Tests	34-36 Tests	Provenance / Derivation Notes
Conductance Range ($G_{\max}/G_{\min}$)	10		47.3		10		Canonical anchored to scalable OPECTs; Zhang 2026 derives from reported max/min current levels.
Effective States (n_{states})	16 (4-bit)		34		16 (4-bit)		Canonical matches general low-precision target. Zhang 2026 anchored to Fig.3h & Supp.Fig.8.
Cycle-to-Cycle Noise (σ_{c2c})	5%		2%		5% (Proportional)		Zhang 2026 value is a proxy estimate from 8-cycle repeatability (Supp.Fig.15). Proportional stress scales $\sigma \propto G $.
Device-to-Device Mismatch (σ_{d2d})	10%		3%		10% (Proportional)		Zhang 2026 uses 3% as a conservative conductance-domain proxy from the reported $\sim 1\%$ V_{th} spread.
Retention Decay (τ_1, τ_2, A_0)	140ms, 610ms, 0.6		Not fitted		140ms, 610ms, 0.6		Canonical dual-exponential fit anchored to Vincze 2026. Zhang 2026 (Fig.2d) is qualitative only.
Nonlinear Asymmetry (NL)	Write 1.0 (LTP), -1.0 (LTD)		Not injected directly		2.0 (LTP), -2.0 (LTD)		Canonical assumes symmetric behavior. Stress tests ($NL = 2$) enforce severe gradient-scaling asymmetry.
Noise Mode	Uniform		Uniform		Proportional		Uniform injects noise normalized to full range. Proportional applies state-dependent noise magnitudes.

Note: The identical values across C2C variations indicate that the Ensemble HAT model is overwhelmingly dominated by the static D2D spatial mismatch rather than the per-forward C2C sampling noise, reinforcing the structural robustness of the learned weight basins. Even under severely pessimistic D2D mismatch assumptions (up to 15%), the model maintains an accuracy well above 80%, far outperforming the 10% collapse of the standard V4 model.

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Table 2: Ensemble HAT accuracy under varied Zhang proxy noise assumptions.

C2C \ D2D	2%	3% (Nominal)	5%	10%	15%
1%	88.59%	88.55%	88.34%	87.32%	84.60%
2% (Nominal)	88.59%	88.53%	88.34%	87.32%	84.60%
5%	88.59%	88.55%	88.34%	87.32%	84.60%
8%	88.59%	88.55%	88.34%	87.32%	84.60%

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